

1. Introduction to TCP/IP

TCP/IP stands for **Transmission Control Protocol / Internet Protocol**, the fundamental communication suite for the Internet. It defines how data is packaged, addressed, transmitted, routed, and received between devices.

Key Concept: TCP/IP is the backbone that enables all web services, including HTTP, HTTPS, FTP, and email protocols.

2. Layers of the TCP/IP Model

TCP/IP uses a **four-layer architecture** (similar to the OSI model). Each layer has a specific function in data transmission.

Layer	Description	Example Protocols
Application Layer	Provides network services to end-users and applications.	HTTP, HTTPS, FTP, SMTP, DNS
Transport Layer	Ensures reliable communication between devices.	TCP, UDP
Internet Layer	Handles addressing, routing, and packet forwarding.	IP, ICMP
Network Access Layer	Defines how data is transmitted over the physical network.	Ethernet, Wi-Fi

Mnemonic: A → T → I → N → (*Application, Transport, Internet, Network Access*)

3. TCP (Transmission Control Protocol)

TCP is a **connection-oriented** protocol that ensures reliable communication between sender and receiver.

Main Features:

- Establishes a connection before data transfer (3-way handshake).
- Guarantees delivery and maintains order of packets.
- Retransmits lost packets.
- Closes the connection after data exchange.

Three-Way Handshake Process:

1. **SYN:** Client sends synchronization request.
2. **SYN-ACK:** Server acknowledges and responds.
3. **ACK:** Client confirms the connection.

Advantages: Reliable and ordered communication (used in HTTP, HTTPS, FTP).

4. UDP (User Datagram Protocol)

UDP is a **connectionless** protocol that sends data without establishing a formal connection.

Main Features:

- No connection setup or acknowledgment.
- Faster but less reliable than TCP.
- Used where speed is preferred over reliability (e.g., video streaming, gaming).

Comparison: TCP vs UDP

Feature	TCP	UDP
Type	Connection-oriented	Connectionless
Reliability	Reliable	Unreliable
Speed	Slower	Faster
Use Case	Web pages, emails	Video, audio, DNS

5. IP (Internet Protocol)

The Internet Protocol defines how packets are addressed and routed between source and destination across the Internet.

Types of IP:

- **IPv4:** 32-bit address (e.g., 192.168.1.1)
- **IPv6:** 128-bit address, supports larger address space.

Functions:

- Logical addressing
 - Packet routing between networks
 - Fragmentation and reassembly
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6. Supporting Protocols in TCP/IP Suite

Protocol	Function
ICMP (Internet Control Message Protocol)	Reports errors and diagnostics (e.g., Ping).
ARP (Address Resolution Protocol)	Maps IP addresses to MAC addresses.
DNS (Domain Name System)	Translates domain names (e.g., google.com) to IP addresses.
DHCP (Dynamic Host Configuration Protocol)	Automatically assigns IP addresses to devices.

7. Application Layer Protocols

These are built on top of TCP/IP and provide services directly to users or applications.

1. HTTP (Hyper Text Transfer Protocol)

- **Purpose:** Transfer of web pages (text, images, video).
- **Operation:** Browser sends request (GET /index.html), and the web server responds with the page.
- **Port:** 80 (HTTP), 443 (HTTPS)
- **Example:** Opening www.google.com

2. FTP (File Transfer Protocol)

- **Purpose:** Uploading and downloading files between client and server.
- **Operation:** User connects with username/password. Files can be transferred, renamed, or deleted.
- **Port:** 20 (data), 21 (control)
- **Example:** Uploading a website to a hosting server.

3. SMTP (Simple Mail Transfer Protocol)

- **Purpose:** Sending email from client to server.
- **Operation:** Mail client sends outgoing mail to mail server. Server forwards it to recipient's mail server.
- **Port:** 25 (SMTP)
- **Example:** Sending an email through Gmail.

4. POP3 (Post Office Protocol v3)

- **Purpose:** Receiving and downloading emails to local computer.
- **Operation:** Connects to mail server, downloads and deletes mail from server (by default).
- **Port:** 110
- **Example:** Outlook downloading emails.

5. IMAP (Internet Message Access Protocol)

- **Purpose:** Receiving and managing emails on the server.
- **Operation:** Emails remain on the server. User can read, organize, and delete from multiple devices.
- **Port:** 143 (IMAP), 993 (IMAP Secure)
- **Example:** Gmail app syncing mail on phone and PC.

6. DNS (Domain Name System)

- **Purpose:** Converts domain names to IP addresses.
- **Operation:** Client asks DNS server: "What is IP of www.example.com?" DNS replies: "93.184.216.34."
- **Port:** 53
- **Example:** Typing www.youtube.com in browser.

7. TELNET

- **Purpose:** Remote login to a computer/server.
- **Operation:** Sends keystrokes to remote machine and executes commands as if user is sitting there.
- **Port:** 23
- **Note:** Insecure; mostly replaced by SSH.

8. SSH (Secure Shell)

- **Purpose:** Secure remote login and command execution.
- **Operation:** Provides encrypted communication with remote servers.
- **Port:** 22
- **Example:** Admin managing a Linux server.

8. Data Encapsulation and Flow

Data in TCP/IP moves through layers in a process called **Encapsulation**:

1. **Application Layer:** Creates data.
2. **Transport Layer:** Adds TCP/UDP header.
3. **Internet Layer:** Adds IP header.
4. **Network Access Layer:** Adds physical addressing and converts to bits.

De-encapsulation occurs in reverse order at the receiving end.

9. Importance of TCP/IP in Web Applications

- Enables global connectivity between servers and browsers.
- Forms the base for HTTP communication.
- Allows web servers to host multiple clients simultaneously.
- Ensures reliable data transmission between distributed systems.

Example: When you open a website, your browser (client) uses TCP/IP to communicate with the server, request data (HTTP), and receive the web page content.

10. Summary and Revision Points

Concept	Key Idea
TCP/IP	Core Internet communication model.
Layers	Application, Transport, Internet, Network Access.
TCP	Reliable, ordered, connection-based communication.
UDP	Fast, connectionless, for streaming and games.
IP	Handles addressing and routing of packets.
DNS	Translates domain names to IP addresses.
Encapsulation	Wrapping data with headers at each layer.
HTTP/FTP/SMTP/IMAP	Core services of Application Layer.

